

STEFANUS YUDI IRWAN

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PROFILE

A results-driven engineer with a unique background in oil and gas field operations and AI product development. Driven by a mission to bridge technological gaps in Indonesia's digital infrastructure, I aim to specialize in applied and efficient AI systems. I am seeking to leverage my holistic perspective to contribute to global technological innovation and develop deployable solutions for real-world engineering challenges.

EDUCATION

Universitas Gadjah Mada

Yogyakarta, Indonesia

Bachelor of Engineering, Engineering Physics

July 2019

Cumulative GPA: 3.82/4.00 (Cum Laude)

- Honors: Ranked #1 in the Engineering Physics graduating class of August 2019
- Relevant Coursework: Computer Programming, Linear Algebra, Probability & Statistics, Numerical Methods, Database System, Automatic Control, Sensor & Actuator, Electronics, Signal Processing
- Thesis: Development of Coal Mill Control Structure Based On IEC 61499

PUBLICATION

Distributed Coal Mill Simulator based on IEC 61499 [\[paper\]](#)

2019

Published in International Conference on Science and Technology (ICST), Yogyakarta, Indonesia, 2019, pp. 1–6. DOI: 10.1109/ICST47872.2019.9166357.

Perancangan Struktur Kontrol Penggilingan Batubara pada Sistem Pembakaran Batubara [\[paper\]](#)

2019

Published in Jurnal Rekayasa Elektrika, vol. 15, no. 3, pp. 177–185, Dec. 2019, DOI: 10.17529/jre.v15i3.14605.

PROJECT EXPERIENCE

Car Detection and Classification [\[Doc\]](#)

June 2025

- Built a computer vision system using YOLOv5 and VGG16 to detect and classify Indonesian car models, achieving 91.2% mAP in detection and 72.5% accuracy in classification on a custom regional dataset.

KG Media, AI Assistant and News Pre-Moderation [\[Doc\]](#)

June 2024 – December 2024

- Engineered an LLM application to automate content preprocessing (paraphrasing, metadata enrichment, pre-moderation), boosting team productivity and increasing daily article publication by around 46.5%.

KG Media, Kompasiana Article Classification [\[Doc\]](#)

September 2024 – October 2024

- Developed a classification system using a fine-tuned BERT model to automatically tag policy violations in user-generated content, enhancing moderation productivity and reducing violations by around 24%.

Pacmann Academy, Neural Collaborative Filtering [\[Doc\]](#)

January 2024

- Built a neural movie recommendation system, benchmarking architectures (GMF, MLP, NeuMF) to achieve Hit Ratio over 65% and NDCG over 38%.

KG Media, Media Recommendation Systems [\[Doc\]](#)

July 2023 – October 2023

- Implemented multi-content recommendation systems (articles, videos, books) using similarity, statistical, and personalized algorithms, increasing user engagement by $\pm 2\%$ and maintaining CTR around 20% for KG Media.

Pacmann Academy, Lithofacies Classification [\[Doc\]](#)

November 2022

- Built a lithofacies classification model by evaluating multiple supervised algorithms on oil well log data, ultimately selecting an XGBoost classifier for achieving the highest performance (86.5% ROC-AUC, 60.8% CV accuracy).

WORKING EXPERIENCE

EY Indonesia <i>Senior Technology Consultant Data & AI (Full Time)</i>	Jakarta, Indonesia January 2025 – Present
Design enterprise integration systems by developing API contracts to seamlessly connect diverse client infrastructure, ensuring technical solutions meet critical business requirements.	
KG Media <i>Data Scientist (Full Time)</i>	Jakarta, Indonesia March 2023 – December 2024
Engineered scalable AI/ML systems using Python, Golang, and cloud infrastructure (GCP, Kubernetes) to power core KG Media products, driving measurable gains in content team productivity and efficiency.	
Artha Dana Teknologi <i>Operation Strategist (Full Time)</i>	Jakarta, Indonesia October 2022 – March 2023
- Analyzed repayment data to optimize collection strategies, targeting 100% repayment while maintaining costs between 2-5%. - Advised managers of three operations divisions, Desk Collection, Quality Assurance, and Mystery Shopper, enhancing service levels and reducing service time by 20%.	
Halliburton Logging Services Indonesia <i>Wireline Field Professional (Full Time)</i>	Balikpapan, Indonesia January 2020 – June 2022
- Supervised a team of 4–5 field personnel in executing data acquisition projects for oil and gas wells across 19 locations in the Borneo Region. - Managed all aspects of field operations, ensuring strict adherence to safety protocols and successful project completion.	
Universitas Gadjah Mada <i>Lecturer Assistant and Laboratory Assistant (Part Time)</i>	Yogyakarta, Indonesia June 2017 – June 2019
- Taught students in electronics, control, and sensor laboratory practices while conducting extra lessons to enhance their understanding. - Authored performance reports to track progress and collaborated with lecturers to review assignments regularly.	

CERTIFICATION [[Certificates](#)]

Recommender System (<i>Pacmann Academy</i>)	April 2024
Natural Language Processing (<i>Deep Learning AI</i>)	March 2024
Deep Learning (<i>Pacmann Academy</i>)	January 2024
Building Machine Learning Product (<i>Pacmann Academy</i>)	July 2023
Advanced Machine Learning (<i>Pacmann Academy</i>)	July 2023
Machine Learning (<i>Deep Learning AI</i>)	May 2023

SKILLS

- AI & Machine Learning:** Supervised & Unsupervised Learning, Natural Language Processing (NLP), Deep Learning, Large Language Models (LLMs), Computer Vision, Recommender Systems
- Programming Languages:** Python, Golang, SQL, R
- Data Tools & Databases:** PostgreSQL, Google BigQuery, Redis, Tableau, Pandas, Matplotlib, Seaborn
- Software & Cloud Infrastructure:** Docker, Kubernetes, CI/CD Pipelines, Google Cloud Platform
- Languages:** Bahasa Indonesia (Native), English (Advanced - IELTS 7.5), German (Beginner)
- Soft Skills:** Problem-Solving, Critical Thinking, Leadership, Project Management, Cross-Functional Collaboration, Adaptability